Residual Corrected Flow for Human-Centric Editing Across Video Shots and 3D Assets

Wenshuo Gao, Jiangyue Zeng, and Shuai Yang, *Member, IEEE,*

***Abstract*—Human-centric editing revises scenes, clothing, props, and illumination while preserving the actor's identity and recorded performance. We present FlowCine, a training-free framework for controllable editing of video shots and native 3D human assets using region-dependent residual-corrected flow. Users specify natural-language instructions and may independently supply background, clothing, and object images. A pretrained image editor synthesizes a target appearance, which guides the video and 3D branches. For video, mixed depth-and-edge control constrains the head, while depth control follows the remaining person's structure and motion. Head-focused residual correction and detail transfer preserve identity-related information as clothing and surroundings change. Shared instructions and visual references extend the interface to separately processed shots with a common design. For 3D, FlowCine encodes a supplied mesh into native shape and physically based appearance latents. Multi-view semantic labels define spatial head mask for residual correction, and coordinate-aware frequency transfer preserves detail on sparse three-dimensional features. The edited latents decode into reusable assets with modified geometry and materials. Experiments on videos and native 3D assets demonstrate controllable appearance editing with preservation of identity, structure, and fine details.**

***Index Terms*—Video editing, flow matching, 3D editing.**

I. Introduction

BRINGING a visual concept to life through physical production is costly. Securing locations, preparing costumes and props, and arranging lighting require substantial resources and coordination, with each visual design imposing its own production demands. The ability to transform these elements in footage captured on a modest budget to meet more ambitious visual and production requirements would let a single recorded performance support a broader range of visual concepts. Human-centric editing offers this flexibility by changing the scene, clothing, objects, and illumination while retaining the performer's identity, details, and motion. For 3D human assets, similar control would make these visual choices adjustable across viewpoints, supporting the reuse of digital performers in virtual production.

Jointly changing setting, clothing, and objects creates a conflict between preservation and editability. Directly applying these edits can change the person's face and fine details, weakening Identity Consistency. Conversely, strong constraints on the source appearance can preserve unwanted clothing textures, object shapes, or background content, preventing the requested changes. This trade-off becomes more challenging in video, where both the preserved details and the edited appearance must remain consistent as the person moves. An effective editing framework must protect identity-related information while allowing the regions involved in the requested edit to change freely.

We present **FlowCine** (Fig. 1), a training-free framework for controllable human-centric editing of videos and native 3D assets. Users specify an edit in natural language and may add background, clothing, and object images independently. A pretrained image editor combines these conditions into a target appearance, and a pretrained generative model transfers that appearance while following the source structure. This separates the choice of visual design from the recorded performance and provides the same interface for text-driven and reference-guided edits.

The central design is *region-dependent preservation*. In video, the head receives mixed depth-and-edge control, residual correction, and detail transfer. The remaining person region receives depth control, allowing clothing color, pattern, and material to change while following the source pose. The background remains purely generative under the supplied conditions. Together, these components balance identity preservation, Motion Consistency, and editing flexibility within one flow-based process.

For native 3D editing, FlowCine encodes a supplied asset into shape and PBR appearance latents. Multi-view semantic labels define a spatial head region, and residual correction operates directly in both latent spaces. Coordinate-aware frequency transfer uses the XYZ locations of sparse features to preserve local detail. The edited latents are decoded into an asset that can be viewed from new directions. For recorded multi-shot applications, shared instructions and optional visual assets provide a common design across shots, while each shot retains its own source composition and motion.

This work extends our previous work, FlowPortal [1], which studied video relighting and background replacement. FlowCine broadens the task to joint scene, clothing, prop, and illumination editing with independently optional visual references. Its technical additions include region-dependent structural control and head preservation for video, and native 3D shape and PBR appearance editing with semantic spatial masks and coordinate-aware frequency transfer. We also introduce shared-design multi-shot applications and expand the experimental evaluation with video and multi-view 3D comparisons, module ablations, and regional structure, detail, and motion metrics.

Our contributions are threefold:

- A unified human-centric editing framework covering

The authors are with the Wangxuan Institute of Computer Technology, Peking University, Beijing, China (e-mail: {gaowenshuo, zengjy}@stu.pku.edu.cn; williamyang@pku.edu.cn).

Corresponding author: Shuai Yang.

Fig. 1. Human-centric video and native 3D editing. FlowCine supports text-guided and reference-guided changes to scenes, clothing, objects, and illumination while preserving human identity, details, and motion. Users can independently supply scene, clothing, and object references, and edit native 3D assets with consistent appearance across viewpoints. Small thumbnails show the supplied reference images; prompt summaries accompany text-only edits.

scene, clothing, prop, and illumination changes, with independently optional visual references and a shared interface for video and native 3D assets.

- A condition consistency formulation linking directional change with reconstruction under identical conditions, and a residual-corrected flow method that transfers source reconstruction corrections to target generation to preserve source content while enabling the requested edits.
- Region-dependent structural control and detail transfer for separating identity-related information from editable appearance, together with coordinate-aware residual correction and frequency transfer in sparse 3D latents.

Section II reviews related work. Sections III and IV present residual-corrected flow for human-centric video editing and its extension to native 3D assets, respectively. Section V reports comparisons and ablation studies, and Section VI demonstrates multi-shot video editing. We discuss limitations and conclude in Sections VII and VIII.

II. Related Work

A. Human-Centric Video Editing

Video editing balances requested changes against source preservation. Inversion-based approaches [2]–[5] connect editing to reconstruction, while FateZero [6], TokenFlow [7], and FRESCO [8] use attention or correspondence constraints for consistent propagation. AnyV2V [9] propagates an image-edited first frame through temporal feature injection, FastVideoEdit [10] uses consistency models for efficient sampling, and FlowEdit [11] constructs inversion-free trajectories with pretrained flow models.

VideoPainter [12] introduces plug-and-play masked-video context control, while VACE [13] unifies source videos, masks, and visual references within one conditioning interface. Kiwi-Edit [14], Lucy Edit 1.1 [15], UniVideo [16], and DITTO [17] provide instruction-guided editing; Kiwi-Edit and UniVideo also accept reference images. Omni-Video 2 [18] integrates multimodal understanding, generation, and editing. VIVA [19] uses vision-language conditioning and reward-based post-training, while EasyV2V [20] studies data construction and flexible input conditioning. Bai et al. [21] adapt image editors to video latents, and VicEdit [22] learns from image, image-pair, and video-pair editing examples alongside text. EditVid [23] combines temporal memory, token injection, and latent blending for training-free text- and reference-guided editing. These methods face a trade-off between editing flexibility and detail preservation. Direct editing often struggles to preserve fine source details, while stronger source constraints can restrict the requested appearance changes.

B. Video Relighting and Background Replacement

Video relighting preserves surface appearance, geometry, and motion while changing illumination. Early approaches include relightable performance capture [24] and real-time neural portrait relighting with temporal consistency modeling [25]. Total Relighting [26] and SwitchLight [27] estimate normals and albedo for lighting control, while Relightful Harmonization [28] and IC-Light [29] learn foreground relighting and illumination harmonization with supplied backgrounds.

RelightVid [30] and Lux Post Facto [31] train video diffusion models for consistent relighting, including portrait perfor-

mances. TC-Light [32] and Lumen [33] address long videos and background replacement, respectively. IllumiCraft [34] and UniRelight [35] couple illumination with geometry and intrinsic decomposition, respectively. LumiSculpt [36] studies consistent portrait lighting, and UniLumos [37] unifies image and video relighting with physics-plausible feedback. Training-free AnyPortal [38] and Light-A-Video [39] combine image relighters with video models through cross-frame attention and refinement projection, and consistent light attention with progressive light fusion, respectively. FlowPortal [1] uses residual-corrected flow for training-free relighting and background replacement.

C. Virtual Try-On and Appearance Conditioning

Virtual try-on transfers clothing while preserving identity and pose. StableVITON [40] and IDM-VTON [41] learn clothing–body correspondence in diffusion latents and inject high-level clothing semantics and low-level visual features through separate attention paths, respectively; CatVTON [42] spatially concatenates person and clothing inputs.

For video, ClothFormer [43], CatV2TON [44], and DPIDM [45] address temporal consistency through clothing warping and temporal appearance-flow modeling, temporal concatenation for image and video try-on, and dynamic pose interaction, respectively. KeyTailor [46] and BooM-VVT [47] use selected keyframe details for clothing fidelity and Motion Consistency; KeyTailor also improves background integrity, while BooM-VVT learns mask-free localization from image-level pseudo data and uses shared positional encoding for frame correspondence. OmniVTON [48] and PG-VTON [49] enable training-free transfer through decoupled clothing/pose constraints and patch-guided reference alignment, respectively. PROMO [50] combines flow matching, multimodal latent conditioning, and self-reference acceleration for efficient try-on.

D. Appearance Editing and Native 3D Priors

Text2Tex [51] and TexFusion [52] use image diffusion for mesh texturing through depth conditioning and shared multi-view texture denoising, respectively. Instruct-NeRF2NeRF [53] alternates image editing with radiance-field optimization, while GaussianEditor [54] uses semantic tracing and hierarchical Gaussian representations for localized scene editing.

TRELLIS [55] and TRELLIS.2 [56] provide structured native latents for geometry and appearance, with rectified-flow generation in TRELLIS and compact geometry and material representations in TRELLIS.2. AnchorFlow [57] aligns source and target flow trajectories with latent anchors for training-free shape editing, and 3D-LATTE [58] performs training-free geometry editing in native 3D diffusion latents. Easy3E [59] combines voxel flow editing with normal-guided multi-view texture refinement. Hunyuan3D 2.1 [60] paints PBR materials on supplied meshes, while Steer3D [61] learns text-controlled image-to-3D editing from synthetic data. Vinedresser3D [62] and Prox-E [63] localize edits through agent-guided latent inpainting and primitive-based shape abstractions, respectively, translating instructions into spatial guidance for native generative models. PriorEdit3D [64] learns feed-forward editing without paired 3D supervision by combining image-editing, semantic, and native 3D priors.

E. Visual Continuity Across Shots

Cross-shot storytelling preserves characters and scene attributes across composition, action, and viewpoint changes. ConsistI2V [65] improves within-clip consistency through first-frame attention and noise initialization. StoryDiffusion [66] and StoryMaker [67] maintain cross-image consistency through consistent self-attention and joint face-identity and character-image conditions, respectively; StoryMaker also retains clothing, hair, and body appearance. StoryMem [68] and OneStory [69] condition successive shots on historical keyframes, using an updated visual memory in StoryMem and adaptive context selection and compression in OneStory. HoloCine [70] generates scenes jointly with shot-specific text and sparse cross-shot attention. STAGE [71] uses per-shot start–end frame storyboards and shared memory for long-range entity consistency, while DreamShot [72] supports text- and reference-conditioned storyboard synthesis and continuation with video diffusion priors.

III. Residual-Corrected Flow for Human-Centric Video Editing

A. Condition Consistency

FlowCine edits a source video X according to a text instruction and optional background, clothing, and object images. A pretrained conditional flow model supplies the generative prior. Let z_t denote a video latent, where $t = 1$ corresponds to Gaussian noise and $t = 0$ to a clean video. Under condition c, the model predicts the denoising velocity

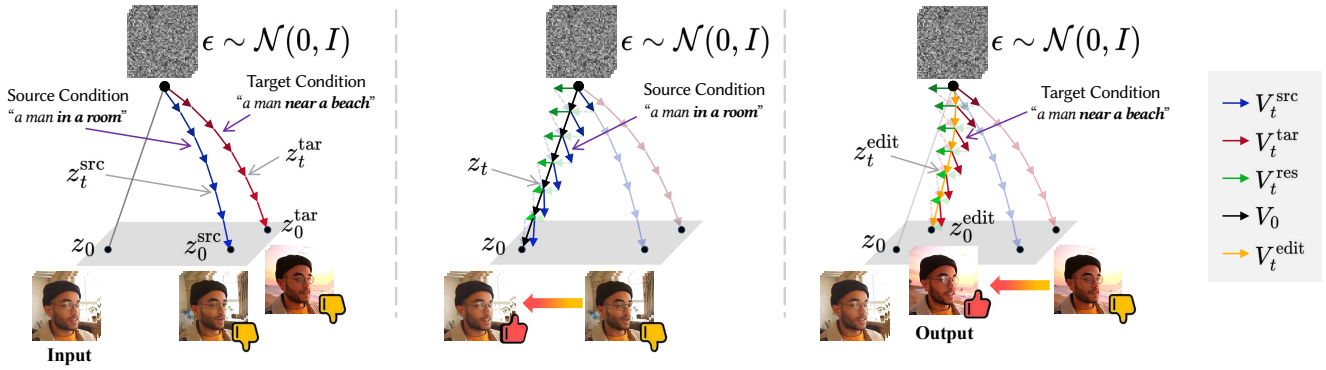
$$V_t^c(z_t) = F_\theta(z_t, t, c), \qquad \frac{\mathrm{d}z_t}{\mathrm{d}t} = -V_t^c(z_t). \tag{1}$$

Generation follows this field from $t = 1$ to $t = 0$. Our aim is to guide this process toward a requested appearance while preserving the identity and performance of the source.

Condition Consistency expresses the relationship between a requested edit and the resulting change in the video. An ideal editor responds to the difference between source and target conditions while retaining attributes that the instruction leaves unchanged. We formulate this requirement through two properties.

a) Directional change: When the source and target conditions differ in a particular attribute, the output should change in that attribute while preserving the remaining content. For example, when the source and target conditions differ only in lighting, the output should change accordingly while preserving structure, motion, and fine details unrelated to illumination.

b) Stability under identity: In the extreme case where the source and target conditions are identical, an ideal editing process should reconstruct the recorded input. A zero condition difference should therefore produce a zero edit. Faithful source reconstruction provides the reference from which a controlled edit can be defined.



(a) Direct Inference (b) Calculate Consistency Residual Velocity (c) Inference Using Residual-Corrected Flow

Fig. 2. **Illustration of Residual-Corrected Flow.** (a) Direct inference builds source- and target-conditioned denoising trajectories from the same noise ϵ. The generated source z_0^{src} differs from the recorded input z_0. (b) The consistency residual velocity V_t^{res} is the difference between the input reconstruction velocity V_0 and the predicted source velocity V_t^{src}, correcting the source trajectory toward the input. (c) Combining the target velocity V_t^{tar} with this residual produces the edited trajectory, preserving source structure while following the target condition. Purple arrows indicate the conditioning used to predict velocities.

Direct conditional generation exposes the connection between these properties. As illustrated in Fig. 2(a), one can start from the same noise and generate z_0^{src} and z_0^{tar} under the source and target conditions. The condition describes the input incompletely and the pretrained model introduces its own reconstruction error. As a result, the source-conditioned output can differ from the recorded latent z_0. Consequently, editing a real recording combines the intended condition change with this source reconstruction discrepancy. The actor's structure or details can drift before the requested edit is accounted for.

Our approach first corrects the source trajectory so that it reconstructs the given recording, then transfers the same correction to the target trajectory. Condition decoupling identifies the attributes to change, and regional preservation assigns stronger source constraints to identity-bearing content. These components connect stability under identity with directional change in human-centric editing.

B. Residual-Corrected Flow

Let z_0 be the encoded source video and ϵ a fixed Gaussian sample. We construct a restoration path from this noise to the observed input:

$$V_0 = z_0 - \epsilon, \qquad z_t = (1-t)z_0 + t\epsilon. \tag{2}$$

Following the constant velocity V_0 from $t = 1$ to $t = 0$ reconstructs z_0. This known path provides a source reconstruction reference at every noise level.

a) Consistency residual velocity: At z_t, the pretrained model predicts the source-conditioned velocity $V_t^{\text{src}}(z_t) = F_\theta(z_t, t, c_{\text{src}})$. Its difference from the reference velocity defines the consistency residual:

$$V_t^{\text{res}}(z_t) = V_0 - V_t^{\text{src}}(z_t). \tag{3}$$

Adding this residual to the source prediction yields V_0, aligning the source-conditioned trajectory with the recorded input, as shown in Fig. 2(b).

b) Transferring the correction to the target flow: We use the same residual to correct the target-conditioned velocity, yielding the residual-corrected flow:

$$V_t^{\text{edit}}(z_t^{\text{edit}}) = V_t^{\text{tar}}(z_t^{\text{edit}}) + V_t^{\text{res}}(z_t). \tag{4}$$

Starting from $z_1^{\text{edit}} = \epsilon$, denoising with this field produces the edited video in Fig. 2(c). The target velocity supplies the desired appearance change, while the residual anchors the process to the observed source. Under identical source and target conditions, their common initialization gives coincident trajectories and $V_t^{\text{src}} + V_t^{\text{res}} = V_0$, recovering the source path. When the target condition changes, the target flow introduces the requested change around this reconstruction reference.

For human-centric editing, identity and editable appearance occupy different regions. FlowCine applies the consistency residual to the head mask H:

$$V_t^{\text{edit}}(z_t^{\text{edit}}) = V_t^{\text{tar}}(z_t^{\text{edit}}) + H \odot V_t^{\text{res}}(z_t). \tag{5}$$

This concentrates source preservation on facial structure and hair while allowing clothing, props, and the surrounding scene to follow the target design while it ensures that the purely generative process for new clothing textures and background structures remains free from interference.

C. Separated Conditions

An image editor converts the instruction p_{edit} and optional visual assets $\mathcal{A}$ into a target appearance:

$$r_{\text{tar}} = G(r_{\text{src}}, p_{\text{edit}}, \mathcal{A}). \tag{6}$$

Here r_{src} is the source first frame, and $\mathcal{A}$ may be empty. We combine text descriptions, image appearance, and shared source structure as

$$c_{\text{src}} = (q_{\text{src}}, r_{\text{src}}, S), \qquad c_{\text{tar}} = (q_{\text{tar}}, r_{\text{tar}}, S). \tag{7}$$

The target text and image specify the new design, while the common structural condition S retains the recorded pose and performance. This separation supports both text-driven and reference-guided editing through the same interface.

Fig. 3. Source frame, person mask, head mask, and regional control. The head uses mixed depth/edge control, the remaining person uses depth, and the background control is zero.

D. Identity-Aware Regional Structural Control

Structural control follows the same division between preservation and editability. The head receives depth and edge information to retain identity-bearing geometry. The remaining person receives depth to follow the recorded pose while allowing new clothing appearance. The background is synthesized under the target setting. With person mask P and head mask H, the control is

$$S = H \odot S_{\text{head}} + (P - H) \odot D, \tag{8}$$

where S_{head} combines depth and edges, and D denotes source depth. Figure 3 illustrates this construction.

E. Fine Detail Preservation using Frequency Transfer

To further improve fine-grained detail fidelity, we introduce high-frequency transfer from the source to the edited video. Let $\mathrm{LF}(u)$ and $\mathrm{HF}(u)$ denote the low- and high-frequency components obtained by Fourier decomposition, with $u = \mathrm{LF}(u) + \mathrm{HF}(u)$. At each denoising time, the source latent z_t supplies fine detail, while the edited latent z_t^{edit} supplies the evolving appearance. We transfer source high frequencies within the head mask:

$$\begin{aligned} z_t^{\text{edit}} \leftarrow\ & (1 - H) \odot z_t^{\text{edit}} \\ & + H \odot \big[\, \mathrm{LF}(z_t^{\text{edit}}) \\ & \quad + \lambda\, \mathrm{HF}(z_t) + (1 - \lambda)\, \mathrm{HF}(z_t^{\text{edit}}) \big]. \end{aligned} \tag{9}$$

The weight $\lambda \in [0, 1]$ controls the proportion of source high-frequency information. Increasing λ strengthens facial and hair detail preservation, while the target's low-frequency component carries the requested appearance. The head mask confines transfer to identity-bearing content, leaving clothing texture and the background under target control.

IV. Residual-Corrected Flow in Native 3D Latents

A. Encoding a Supplied Asset

The 3D branch edits an existing textured human asset through native sparse shape and appearance representations. Encoding produces source features z_0^{sh} and z_0^{tex} on a common coarse coordinate set $\mathcal{C}$. The source coordinate set is retained while the feature values change, allowing the decoded asset's geometry and material appearance to follow the requested edit.

B. Appearance Conditions and a Spatial Head Mask

Source and target views provide appearance conditions c_{src} and c_{tar} for the native flow models. The image editor constructs the desired appearance from a source view, the edit instruction, and an optional clothing reference. Multi-view

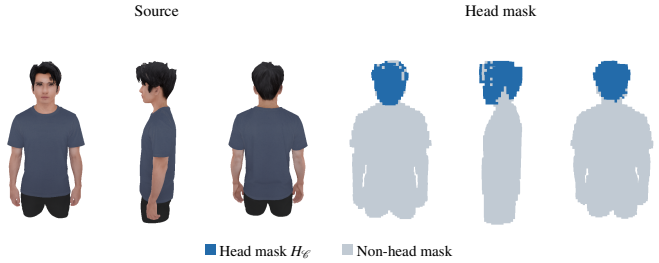


Fig. 4. Spatial head mask for native 3D editing. Multiple views show the source asset and its corresponding sparse latent head mask. Blue marks the protected head mask $H_{\mathcal{C}}$; gray marks the non-head mask.

source semantic labels are lifted onto the observed surface and associated with sparse latent coordinates, defining a spatial head mask $H_{\mathcal{C}}$. This mask assigns identity preservation consistently across viewing directions while leaving the remaining shape and appearance features editable. Figure 4 shows this spatial mask across views.

C. Masked Residual-Corrected Flow

The same reconstruction principle applies to both shape and appearance latents. Let z_0 denote either source representation and ϵ shared Gaussian noise. The reference path and its denoising velocity are

$$V_0 = z_0 - \epsilon, \qquad z_t = (1 - t)z_0 + t\epsilon. \tag{10}$$

The source- and target-conditioned velocities are denoted by $V_t^{\text{src}}(z_t) = F_\theta(z_t, t, c_{\text{src}})$ and $V_t^{\text{tar}}(z_t^{\text{edit}}) = F_\theta(z_t^{\text{edit}}, t, c_{\text{tar}})$. The consistency residual corrects the source-conditioned prediction toward this known path:

$$V_t^{\text{res}}(z_t) = V_0 - V_t^{\text{src}}(z_t). \tag{11}$$

Using the spatial head mask $H_{\mathcal{C}}$, the edited field becomes

$$V_t^{\text{edit}}(z_t^{\text{edit}}) = V_t^{\text{tar}}(z_t^{\text{edit}}) + H_{\mathcal{C}} \odot V_t^{\text{res}}(z_t). \tag{12}$$

Spatial correction preserves identity-bearing features in native coordinates, while the target condition guides changes in shape and material appearance. Both stages share source-derived head mask.

D. Coordinate-Aware Frequency Transfer

Sparse feature rows have no meaningful one-dimensional frequency order. We scatter each feature channel into a three-dimensional grid at its XYZ coordinates, and filter it together with the occupancy mask O. For Gaussian low-pass operator L_b,

$$\mathrm{LP}_{b,\mathcal{C}}(D) = \left.\frac{L_b(OD)}{L_b(O)}\right|_{\mathcal{C}}. \tag{13}$$

Normalizing by filtered occupancy preserves constants at occupied coordinates. Filtered features are gathered back at the original coordinates. With retained high-frequency fraction η, the transfer is

$$\mathcal{T}(D) = \mathrm{LP}_{b,\mathcal{C}}(D) + \eta\big(D - \mathrm{LP}_{b,\mathcal{C}}(D)\big). \tag{14}$$

The bandwidth b and retained fraction η determine the balance between source detail and the target appearance.

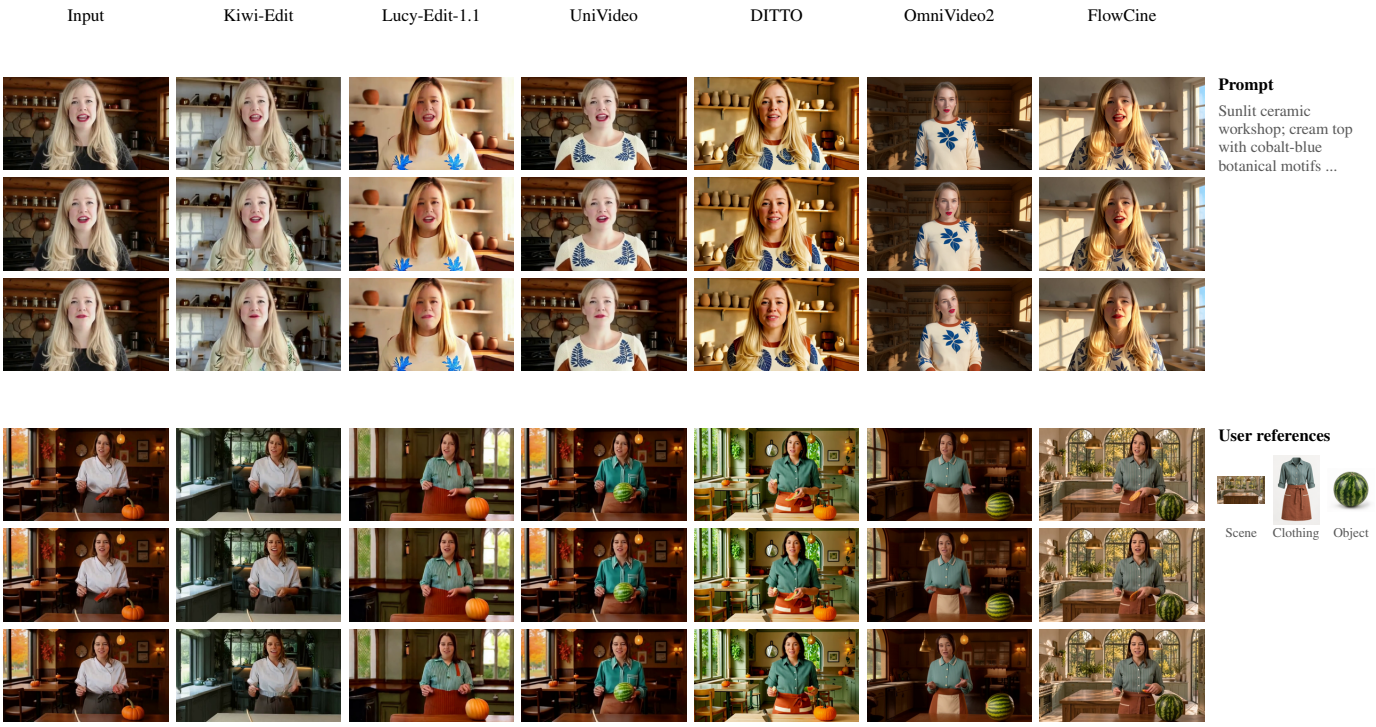


Fig. 5. Qualitive video comparisons.

TABLE I
QUANTITATIVE VIDEO EDITING COMPARISON.

Method	SC ↑	DC ↑	MC ↑	IC ↑	PA ↑	AQ ↑
Kiwi-Edit	0.7435	59.20	0.9883	0.327	0.298	5.148
Lucy-Edit-1.1	0.8126	61.37	0.9866	0.594	0.293	4.968
UniVideo	0.7718	59.73	0.9912	0.704	0.304	4.947
DITTO	0.6270	56.87	0.9883	0.269	0.287	5.100
OmniVideo2	0.7418	58.60	0.9532	0.333	0.303	5.188
FlowCine(Ours)	**0.8310**	**61.89**	**0.9934**	**0.719**	**0.305**	**5.325**

E. Shape-Conditioned Appearance

The shape stage determines the edited geometry, which then conditions appearance editing. Source geometry accompanies the source appearance branch, while edited geometry accompanies the target branch. This coupling adapts the material appearance to the new shape at shared sparse coordinates. Decoding the edited representations produces a textured asset that can be rendered from new views.

V. EXPERIMENTAL RESULTS

A. Implementation Details

a) Evaluation sets: We constructed an video evaluation set contains 64 cases from 48 distinct video clips with 49 frames at different resolutions. Meanwhile, native 3D evaluation set contains 24 supplied human assets.

b) Image and video backbones: The image editor is FLUX.2-klein-9B [73]. It receives the source first frame and enabled optional background, clothing, and object references, identified by role in the instruction. The video branch uses Wan2.1-Fun-V1.1-1.3B-Control [74], [75] with 50 steps.

c) Masks and structural control: FASHN Human Parser [76] initializes person and face/hair masks. Two independent MatAnyone [77] passes propagate person and head masks. All masks and detector maps are resized to the video grid; the head is clipped to the person mask. Head control combines depth, Canny, and HED with weights 0.8, 0.1, and 0.1. The remaining person receives depth, and background control is zero.

d) Video residual correction: Both source and target CFG scales are set to 6. Frequency transfer uses frequency threshold 0.8 with source high-frequency weight $\lambda = 0.5$ in Eq. (9). To accelerate denoising, we refresh the source prediction every 10 steps and reuse the corresponding residual for the intervening steps. The 50-step process therefore requires 50 target and only 5 source predictions, only $1.1\times$ the flow evaluations of pure generation under the same CFG, while preserving facial identity and fine details.

e) Native 3D implementation: TRELLIS.2 [56] runs at resolution 1,024, with 24 shape and 24 appearance steps. The source mesh is normalized to the encoder cube and voxelized with material attributes. Shape and appearance features are matched by exact coordinate keys; a bounded nearest-cell remap handles geometry/material voxelization at source triangle boundaries. Four source views use quarter-turn azimuths, 512×768 resolution, and a 35° horizontal field of view. The frontal view supplies the image editor. Source and target images use a shared source-derived crop and independent person masks. The native flow models receive image conditions. Source face/hair surface labels are mapped to latent cells through the nearest observed source surface points. Coordinate-aware filtering uses blur 0.2, high-frequency fraction 0.5. The appearance model concatenates

Fig. 6. Video module ablation.

TABLE II
VIDEO MODULE ABLATION STUDY.

Variant	SC ↑	DC ↑	MC ↑	IC ↑	PA ↑	AQ ↑
No residual correction	0.6509	57.1364	0.9947	0.3071	0.260	5.428
Full video correction	0.6384	**59.2576**	**0.9981**	**0.6825**	0.244	5.487
No frequency transfer	0.6119	58.6199	0.9971	0.6087	0.279	5.496
Full	**0.6580**	59.1697	0.9972	0.6705	**0.282**	**5.506**

TABLE III
QUANTITATIVE NATIVE 3D EDITING COMPARISON.

Method	SC ↑	DC ↑	IC ↑	PA ↑	AQ ↑
Hunyuan2.1	0.707	50.234	0.402	0.313	4.603
AnchorFlow + PBR	0.715	47.039	0.252	0.268	4.313
FlowCine(Ours)	**0.889**	**60.431**	**0.768**	**0.315**	**4.663**

source or edited shape features with the corresponding appearance branch. Decoded assets are exported as GLBs in the source frame with baked 2048×2048 PBR textures.

B. Evaluation Metrics

Prompt Alignment (PA) measures CLIP cosine similarity between edited frames and the target prompt [6]. **Identity Consistency (IC)** measures ArcFace feature cosine similarity between corresponding source and edited faces [78]. **Aesthetic Quality (AQ)** uses the frozen LAION aesthetic predictor V2 [79]. Regional preservation measures structure, detail, and motion inside common source-defined masks. **Structure Consistency (SC)** is Canny-map SSIM within the source head mask. **Detail Consistency (DC)** is PSNR between LoG responses of albedo predictions [80] on that region, using the frozen Intrinsic v2.0 model for source and output frames. **Motion Consistency (MC)** is SSIM between source and edited RAFT [81] flow components.

TABLE IV
NATIVE 3D MODULE ABLATION STUDY.

Variant	SC ↑	DC ↑	IC ↑	PA ↑	AQ ↑
No residual correction	0.804	54.451	0.276	0.299	4.579
Full-person correction	0.893	59.205	**0.759**	0.308	4.648
No frequency transfer	0.883	58.037	0.645	0.305	4.701
Full	**0.895**	**59.882**	0.758	**0.310**	**4.706**

C. Video Comparisons

We compare Kiwi-Edit [14], Lucy Edit 1.1 [15], UniVideo [16], DITTO [17], and OmniVideo2 [18].

Table I reports the comparison with other methods. FlowCine obtains the highest scores on all metrics, indicating the best overall balance between Prompt Alignment, source preservation, and visual quality. Figure 5 compares the source and method outputs, with the requested prompt and supplied user references alongside each example. FlowCine best preserves facial details while faithfully following the requested edits with the best visual quality.

D. Video Module Ablation

We compare FlowCine with variants that remove residual correction, apply correction to the full video, or remove frequency transfer, while keeping the generation settings fixed. Table II shows that residual correction improves structure, motion, and Identity Consistency, while frequency transfer further strengthens fine detail preservation. Figure 6 shows that removing residual correction changes facial appearance and expressions, while removing frequency transfer weakens facial and hair details. Full video correction introduces artifacts in the edited clothing and background. FlowCine preserves facial details and recorded motion while allowing the requested appearance changes.

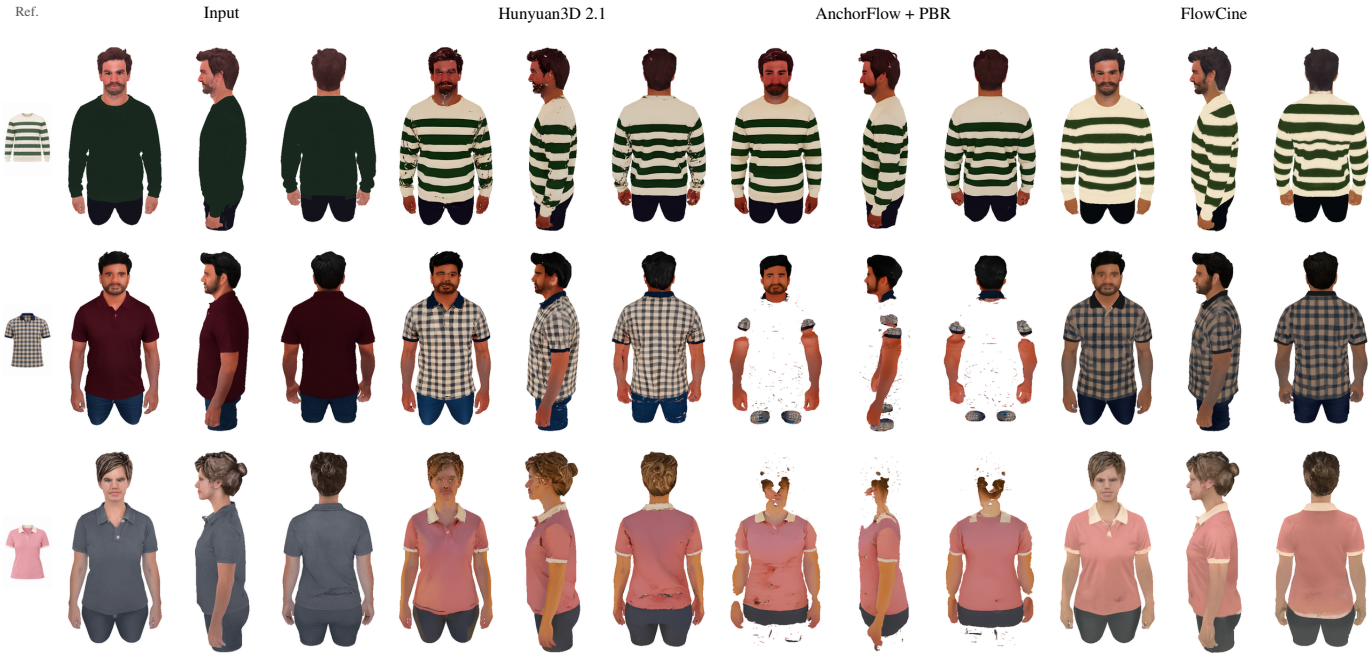


Fig. 7. Qualitative native 3D comparisons.

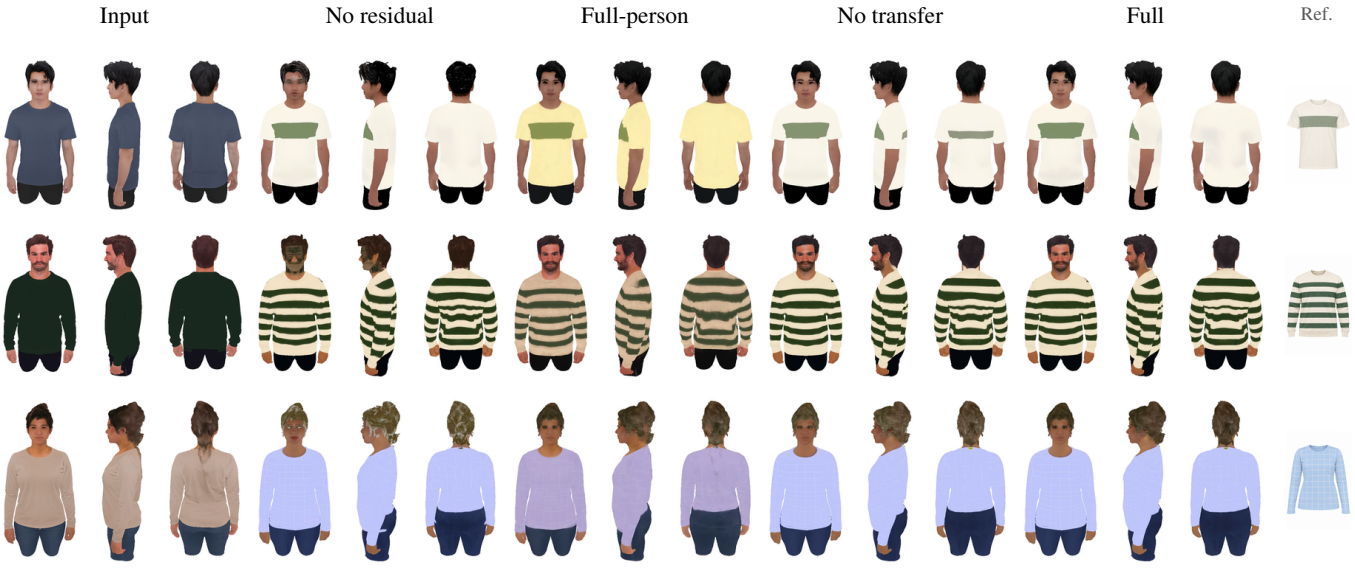


Fig. 8. Native 3D module ablation.

Fig. 9. Multi-shot video editing with scene, clothing, and object references. Source and edited frames are shown for selected shots. Scene, clothing, and object thumbnails show the common landscape, gingham-shirt, and golden-apple conditions.

E. Native 3D Comparisons

We compare Hunyuan3D 2.1 [60] and AnchorFlow [57] with Hunyuan PBR painting.

Table III shows that FlowCine achieves the highest scores across all metrics, combining strong structure, detail, and identity preservation with Prompt Alignment and Aesthetic Quality. Figure 7 compares the source and edited assets across viewpoints. FlowCine preserves facial appearance while applying the requested clothing changes consistently across views.

F. Native 3D Module Ablation

We compare FlowCine with variants that remove residual correction, extend correction to the full person, or remove frequency transfer, while keeping the generation settings fixed. Table IV shows that residual correction and frequency transfer improve structure and detail preservation, while the full model achieves the highest Aesthetic Quality. Figure 8 illustrates how the variants affect facial details and clothing appearance. FlowCine balances source preservation with the requested appearance changes.

VI. Application: Multi-Shot Video Editing

A coherent visual design is essential when edited shots are assembled into a sequence. Changes to the setting, clothing, and props should remain recognizable across camera angles and compositions, while each shot retains the actor's identity and recorded performance. FlowCine supports this workflow by extending its editing interface from individual shots to a shared design across shots.

Users specify a common edit instruction and may supply background, clothing, and object references independently. Each shot generates its own target appearance from these shared conditions and its source frame, adapting the design to the local pose and composition. Region-dependent structural control follows the recorded performance, while head-focused residual correction and detail transfer preserve identity-related features. Shots are processed independently, allowing the same design to be applied without requiring aligned frames or matching camera viewpoints.

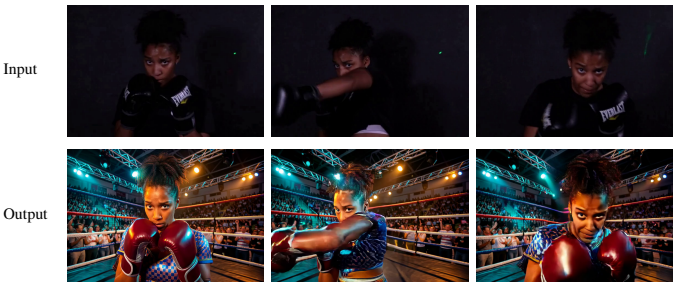


Fig. 10. Limitations on challenging video edits.

Figure 9 illustrates coordinated changes to the scene, clothing, and props across shots. The edited outputs maintain a recognizable shared setting and wardrobe while accommodating changes in body orientation, framing, and hand position. Facial appearance and recorded gestures remain recognizable as the surrounding visual design changes. These results demonstrate FlowCine's usefulness for coordinated restyling of recorded footage, with shared references providing visual guidance across independently edited shots.

VII. Limitations

FlowCine is constrained by the capabilities of the underlying image editing and video generation models. As shown in Fig. 10, it struggles to produce fine details when complex backgrounds, lighting, motion, and clothing are combined.

VIII. Conclusion

We presented FlowCine for controllable human-centric editing of videos and native 3D assets. Region-dependent

structural control, residual correction, and detail transfer coordinate head preservation with scene, clothing, prop, and illumination changes. Text and independently optional visual references provide a common interface. The native branch edits shape and PBR latents using semantic spatial masks and coordinate-aware frequency transfer. Single-shot experiments, multi-shot applications, and multi-view comparisons examine preservation and requested appearance changes. Future work will develop joint cross-shot scene modeling, stronger object–hand interaction control, and material editing under varying illumination.

REFERENCES

[1] W. Gao, J. Fan, J. Zeng, and S. Yang, "FlowPortal: Residual-corrected flow for training-free video relighting and background replacement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 2025–2034.

[2] J. Song, C. Meng, and S. Ermon, "Denoising diffusion implicit models," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021.

[3] A. Hertz, R. Mokady, J. Tenenbaum, K. Aberman, Y. Pritch, and D. Cohen-Or, "Prompt-to-Prompt image editing with cross-attention control," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023.

[4] R. Mokady, A. Hertz, K. Aberman, Y. Pritch, and D. Cohen-Or, "Null-text inversion for editing real images using guided diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 6038–6047.

[5] C. H. Wu and F. De la Torre, "A latent space of stochastic diffusion models for zero-shot image editing and guidance," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 7378–7387.

[6] C. Qi *et al.*, "FateZero: Fusing attentions for zero-shot text-based video editing," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 15 932–15 942.

[7] M. Geyer, O. Bar-Tal, S. Bagon, and T. Dekel, "TokenFlow: Consistent diffusion features for consistent video editing," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024.

[8] S. Yang, Y. Zhou, Z. Liu, and C. C. Loy, "FRESCO: Spatial-temporal correspondence for zero-shot video translation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 8703–8712.

[9] M. Ku, C. Wei, W. Ren, H. Yang, and W. Chen, "AnyV2V: A tuning-free framework for any video-to-video editing tasks," *Trans. Mach. Learn. Res.*, 2024.

[10] Y. Zhang, X. Ju, and J. J. Clark, "FastVideoEdit: Leveraging consistency models for efficient text-to-video editing," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, 2025, pp. 3657–3666.

[11] V. Kulikov, M. Kleiner, I. Huberman-Spiegelglas, and T. Michaeli, "FlowEdit: Inversion-free text-based editing using pre-trained flow models," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2025, pp. 19 721–19 730.

[12] Y. Bian *et al.*, "VideoPainter: Any-length video inpainting and editing with plug-and-play context control," in *Proc. ACM SIGGRAPH Conf. Papers*, 2025, pp. 1–12.

[13] Z. Jiang, Z. Han, C. Mao, J. Zhang, Y. Pan, and Y. Liu, "VACE: All-in-one video creation and editing," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2025, pp. 17 191–17 202.

[14] Y. Lin, G. Liang, Z. Zeng, Z. Bai, Y. Chen, and M. Z. Shou, "Kiwi-Edit: Versatile video editing via instruction and reference guidance," *arXiv preprint arXiv:2603.02175*, 2026.

[15] DecartAI Team, "Lucy Edit: Open-weight text-guided video editing," Technical report; Lucy Edit 1.1 Dev checkpoint. https://huggingface.co/decart-ai/Lucy-Edit-1.1-Dev, 2025, accessed: Oct. 20, 2026.

[16] C. Wei *et al.*, "UniVideo: Unified understanding, generation, and editing for videos," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2026.

[17] Q. Bai *et al.*, "Scaling instruction-based video editing with a high-quality synthetic dataset," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 37 971–37 981.

[18] H. Yang *et al.*, "Omni-Video 2: Scaling MLLM-conditioned diffusion for unified video generation and editing," *arXiv preprint arXiv:2602.08820*, 2026.

[19] X. Cong *et al.*, "VIVA: VLM-guided instruction-based video editing with reward optimization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 34 364–34 374.

[20] J. Mai *et al.*, "EasyV2V: A high-quality instruction-based video editing framework," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 30 435–30 445.

[21] Y. Bai, Y. Gandelsman, M. Gharbi, and Q. Huang, "Instruction-based video editing by repurposing an image editing model," *arXiv preprint arXiv:2608.14790*, 2026.

[22] Y. Wang *et al.*, "VicEdit: Learning to edit videos from visual in-context examples," *arXiv preprint arXiv:2608.16745*, 2026.

[23] A. S. Juvekar *et al.*, "One editor, many edits: A unified training-free framework for diverse video editing," *arXiv preprint arXiv:2609.04190*, 2026.

[24] K. Guo *et al.*, "The relightables: Volumetric performance capture of humans with realistic relighting," *ACM Trans. Graph.*, vol. 38, no. 6, pp. 217:1–217:19, 2019.

[25] L. Zhang, Q. Zhang, M. Wu, J. Yu, and L. Xu, "Neural video portrait relighting in real-time via consistency modeling," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 802–812.

[26] R. Pandey *et al.*, "Total Relighting: learning to relight portraits for background replacement," *ACM Trans. Graph.*, vol. 40, no. 4, pp. 43:1–43:21, 2021.

[27] H. Kim, M. Jang, W. Yoon, J. Lee, D. Na, and S. Woo, "SwitchLight: Co-design of physics-driven architecture and pre-training framework for human portrait relighting," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 25 096–25 106.

[28] M. Ren *et al.*, "Relightful Harmonization: Lighting-aware portrait background replacement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 6452–6462.

[29] L. Zhang, A. Rao, and M. Agrawala, "Scaling in-the-wild training for diffusion-based illumination harmonization and editing by imposing consistent light transport," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025.

[30] Y. Fang *et al.*, "RelightVid: Temporal-consistent diffusion model for video relighting," *arXiv preprint arXiv:2501.16330*, 2025.

[31] Y. Mei *et al.*, "Lux post facto: Learning portrait performance relighting with conditional video diffusion and a hybrid dataset," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2025, pp. 5510–5522.

[32] Y. Liu *et al.*, "TC-Light: Temporally coherent generative rendering for realistic world transfer," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 38, 2025, pp. 51 463–51 489.

[33] J. Zeng *et al.*, "Lumen: Consistent video relighting and harmonious background replacement with video generative models," *arXiv preprint arXiv:2508.12945*, 2025.

[34] Y. Lin, Y.-W. Chen, Y.-H. Tsai, R. Clark, and M.-H. Yang, "IllumiCraft: Unified geometry and illumination diffusion for controllable video generation," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 38, 2025, pp. 27 798–27 829.

[35] K. He *et al.*, "UniRelight: Learning joint decomposition and synthesis for video relighting," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 38, 2025, pp. 108 435–108 459.

[36] Y. Zhang *et al.*, "LumiSculpt: Enabling consistent portrait lighting in video generation," *arXiv preprint arXiv:2410.22979*, 2024.

[37] P. Liu *et al.*, "UniLumos: Fast and unified image and video relighting with physics-plausible feedback," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 38, 2025, pp. 82 052–82 080.

[38] W. Gao, X. Lan, and S. Yang, "AnyPortal: Zero-shot consistent video background replacement," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2025, pp. 18 990–18 999.

[39] Y. Zhou *et al.*, "Light-A-Video: Training-free video relighting via progressive light fusion," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2025, pp. 13 315–13 325.

[40] J. Kim, G. Gu, M. Park, S. Park, and J. Choo, "StableVITON: Learning semantic correspondence with latent diffusion model for virtual try-on," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 8176–8185.

[41] Y. Choi, S. Kwak, K. Lee, H. Choi, and J. Shin, "Improving diffusion models for authentic virtual try-on in the wild," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2024, pp. 206–235.

[42] Z. Chong *et al.*, "CatVTON: Concatenation is all you need for virtual try-on with diffusion models," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025.

[43] J. Jiang, T. Wang, H. Yan, and J. Liu, "ClothFormer: Taming video virtual try-on in all module," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 10 799–10 808.

[44] Z. Chong *et al.*, "CatV2TON: Taming diffusion transformers for vision-based virtual try-on with temporal concatenation," *arXiv preprint arXiv:2501.11325*, 2025.

[45] D. Li *et al.*, "Pursuing temporal-consistent video virtual try-on via dynamic pose interaction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2025, pp. 22 648–22 657.

[46] Q. He *et al.*, "The devil is in the details: Enhancing video virtual try-on via keyframe-driven details injection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 9182–9191.

[47] W. Zhang *et al.*, "BooM-VVT: Boosting mask-free video virtual try-on with image-level pseudo data," in *Proc. ACM Int. Conf. Multimedia*, 2026, to appear.

[48] Z. Yang *et al.*, "OmniVTON: Training-free universal virtual try-on," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2025, pp. 16 702–16 711.

[49] G. Zhao and Y. Peng, "PG-VTON: Single-pass training-free virtual try-on via patch-guided reference alignment," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 7859–7868.

[50] H. Chen *et al.*, "PROMO: Promptable outfitting for efficient high-fidelity virtual try-on," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 16 074–16 084.

[51] D. Z. Chen, Y. Siddiqui, H.-Y. Lee, S. Tulyakov, and M. Nießner, "Text2Tex: Text-driven texture synthesis via diffusion models," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 18 558–18 568.

[52] T. Cao, K. Kreis, S. Fidler, N. Sharp, and K. Yin, "TexFusion: Synthesizing 3D textures with text-guided image diffusion models," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 4169–4181.

[53] A. Haque, M. Tancik, A. A. Efros, A. Holynski, and A. Kanazawa, "Instruct-NeRF2NeRF: Editing 3D scenes with instructions," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 19 740–19 750.

[54] Y. Chen *et al.*, "GaussianEditor: Swift and controllable 3D editing with gaussian splatting," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 21 476–21 485.

[55] J. Xiang *et al.*, "Structured 3D latents for scalable and versatile 3D generation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2025, pp. 21 469–21 480.

[56] ——, "Native and compact structured latents for 3D generation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 14 419–14 429.

[57] Z. Zhou *et al.*, "AnchorFlow: Training-free 3D editing via latent anchor-aligned flows," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 14 387–14 397.

[58] M. Parelli, M. Oechsle, M. Niemeyer, F. Tombari, and A. Geiger, "3D-LATTE: Latent space 3D editing from textual instructions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 14 377–14 386.

[59] S. Hu, Y. Wei, F. Zha, Y. Guo, and J. Zhang, "Easy3E: Feed-forward 3D asset editing via rectified voxel flow," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 12 730–12 740.

[60] Team Hunyuan3D *et al.*, "Hunyuan3D 2.1: From images to high-fidelity 3D assets with production-ready PBR material," *arXiv preprint arXiv:2506.15442*, 2025.

[61] Z. Ma, H. Chen, Y. Yue, and G. Gkioxari, "Feedforward 3D editing via text-steerable image-to-3D," *arXiv preprint arXiv:2512.13678*, 2025.

[62] Y. Chi, X. Li, Z. Huang, and J. M. Rehg, "Vinedresser3D: Towards agentic text-guided 3D editing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 12 673–12 683.

[63] E. Sella, H. Phung, N. Amiel, O. Litany, O. Patashnik, and H. Averbuch-Elor, "Prox-E: Fine-grained 3D shape editing via primitive-based abstractions," in *Proc. ACM SIGGRAPH Conf. Papers*, 2026, pp. 1–12.

[64] H. Wen *et al.*, "Learning 3D editing without paired supervision via generative prior distillation," *arXiv preprint arXiv:2609.04942*, 2026.

[65] W. Ren *et al.*, "ConsistI2V: Enhancing visual consistency for image-to-video generation," *Trans. Mach. Learn. Res.*, 2024.

[66] Y. Zhou, D. Zhou, M.-M. Cheng, J. Feng, and Q. Hou, "StoryDiffusion: Consistent self-attention for long-range image and video generation," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 37, 2024, pp. 110 315–110 340.

[67] Z. Zhou, J. Li, H. Li, N. Chen, and X. Tang, "StoryMaker: Towards holistic consistent characters in text-to-image generation," *arXiv preprint arXiv:2409.12576*, 2024.

[68] K. Zhang *et al.*, "StoryMem: Multi-shot long video storytelling with memory," *arXiv preprint arXiv:2512.19539*, 2025.

[69] Z. An *et al.*, "OneStory: Coherent multi-shot video generation with adaptive memory," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 16 173–16 184.

[70] Y. Meng *et al.*, "HoloCine: Holistic generation of cinematic multi-shot long video narratives," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 461–471.

[71] P. Zhang, Z. Jia, K. Liu, S. Weng, S. Li, and B. Shi, "STAGE: Storyboard-anchored generation for cinematic multi-shot narrative," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 659–669.

[72] J. Huang *et al.*, "DreamShot: Personalized storyboard synthesis with video diffusion prior," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 36 736–36 746.

[73] Black Forest Labs, "FLUX.2 [klein] 9B," Hugging Face model card. https://huggingface.co/black-forest-labs/FLUX.2-klein-9B, 2026, accessed: Oct. 20, 2026.

[74] Team Wan *et al.*, "Wan: Open and advanced large-scale video generative models," *arXiv preprint arXiv:2503.20314*, 2025.

[75] AIGC-Apps, "VideoX-Fun," GitHub repository. https://github.com/aigc-apps/VideoX-Fun, 2025, accessed: Oct. 20, 2026.

[76] FASHN AI, "FASHN human parser: SegFormer for fashion human parsing," Hugging Face model card. https://huggingface.co/fashn-ai/fashn-human-parser, 2024, accessed: Oct. 20, 2026.

[77] P. Yang, S. Zhou, J. Zhao, Q. Tao, and C. C. Loy, "MatAnyone: Stable video matting with consistent memory propagation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2025, pp. 7299–7308.

[78] J. Deng, J. Guo, N. Xue, and S. Zafeiriou, "ArcFace: Additive angular margin loss for deep face recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 4690–4699.

[79] C. Schuhmann, "LAION aesthetics predictor V2," GitHub repository. https://github.com/christophschuhmann/improved-aesthetic-predictor, 2022, accessed: Oct. 20, 2026.

[80] C. Careaga and Y. Aksoy, "Colorful diffuse intrinsic image decomposition in the wild," *ACM Trans. Graph.*, vol. 43, no. 6, pp. 178:1–178:12, 2024.

[81] Z. Teed and J. Deng, "RAFT: Recurrent all-pairs field transforms for optical flow," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2020, pp. 402–419.